

Global vs. s-t Vertex Connectivity Beyond Sequential: Almost-Perfect Reductions & Near-Optimal Separations

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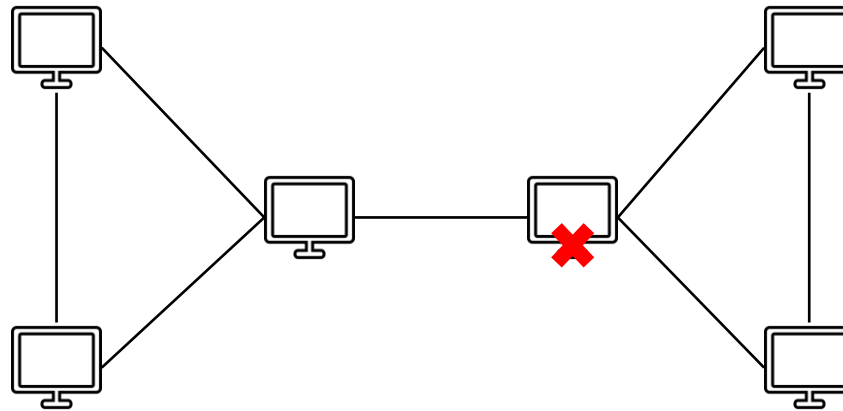
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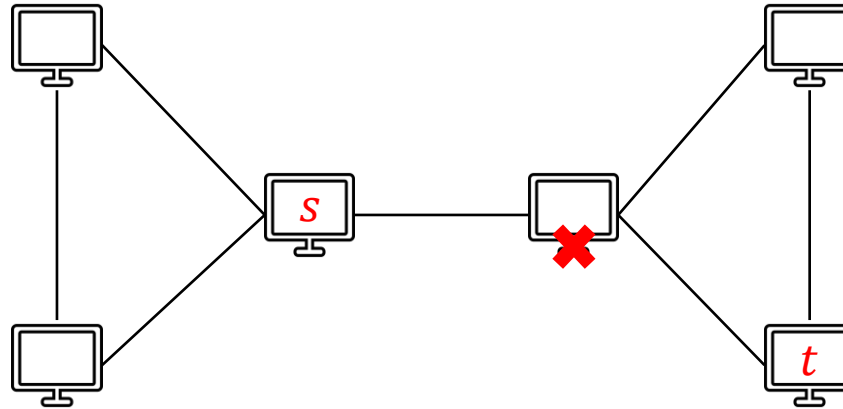
(Global) Vertex Connectivity

- Find the minimum number of **vertices** whose deletion disconnects an **undirected** graph.



s-t Vertex Connectivity

- Find the minimum number of **vertices** whose deletion disconnects **s** and **t** (not adjacent).
- (Folklore) Equivalent to maximum bipartite matching.



How to solve **Global** Vertex Connectivity?

- Trivial: **Global** reduces to $O(n^2)$ calls to **s-t**
 - Recent progress: **Global** reduces to $O(\log^6 n)$ calls to **s-t** [LNPSY'21]

*Is **s-t** necessary for solving **Global**?*

- [HRG'96] [Gabow'00] [LNPSY'21] ... Reductions to **s-t**
- [LLW'86] Convex embedding
- [FNYSY'20] [YS'22] [Korhonen'25] only for small vertex connectivity

How to solve **Global** Vertex Connectivity?

- Trivial: **Global** reduces to $O(n^2)$ calls to **s-t**
- ✓ Recent progress: **Global** reduces to $O(\log^6 n)$ calls to **s-t** [LNPSY'21]

*Is **s-t** necessary for solving **Global**?*

(partially) Yes

- **Our results:** **s-t** reduces to **Global** when graph is dense (has n^2) edges.
- **In contrast:** Is **s-t edge** connectivity necessary for solving **Global edge** connectivity?

(probably) No

- **Global (min-cut):** $O(m \log^3 n)$ [Karger'96] by tree-packing
- **s-t (max flow):** $m^{1+o(1)}$ [CKLPGS'22] by IPM

Is **s-t** sufficient to solve **Global**?

- Sequential: **Global** reduces to $O(\log^6 n)$ calls to **s-t** [LNPSY'21]

Our Results	Parallel (PRAM)	Distributed (CONGEST)	Communication
Reductions	$n^{o(1)}$ calls	$n^{o(1)}$ calls	$n^{0.5}$ calls (tight)
Implications	$n^{\omega+o(1)}$ work, $n^{o(1)}$ depth	$n^{o(1)} \cdot (n^{\frac{3}{8}} D^{\frac{1}{2}} + n^{\frac{1}{2}} D + n^{\frac{7}{10}} D^{7/10})$ (sublinear when $D < n^{\frac{3}{7}}$)	$\tilde{\Theta}(n^{1.5})$ Remark: s-t $\tilde{\Theta}(n)$

Barriers for implementing the sequential algorithm:

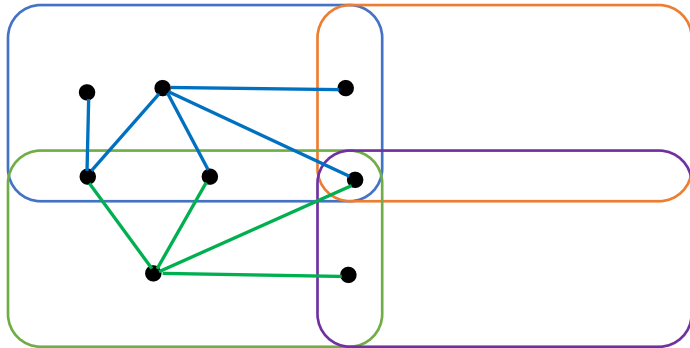
- [LNPSY'21] does not guarantee parallelism
- [LNPSY'21] reduces to **s-t** instances of m edges and n^2 vertices

PRAM, CONGEST Framework (from [LNPSY'21])

- Isolating cuts
 - Parallel (good)
 - ✓ Reduce to m edges and n^2 vertices (our contribution 1: reduce to n vertices)
- Sparsification
 - Not parallel (our contribution 2: use common-neighborhood clustering to parallelize)
 - We give parallel and distributed common-neighborhood clustering
 - Reduce to m edges and n^2 vertices (our contribution 3: reduce to n vertices)
 - Model-specific

Isolating Cuts

Isolating cuts: partition the graph into **edge disjoint** subgraphs, but can share **boundary vertices**

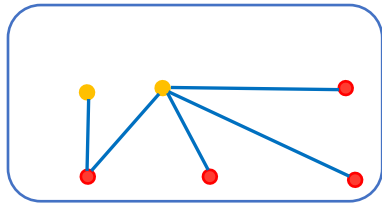


Isolating Cuts

Isolating cuts: split the graph into **edge disjoint** subgraphs, but can share **boundary vertices**

We show for each subgraph, it suffices to find a maximum matching between **inner** and **boundary** vertices, and keep only the matched **boundary** vertices.

-> total vertices becomes $O(n)$

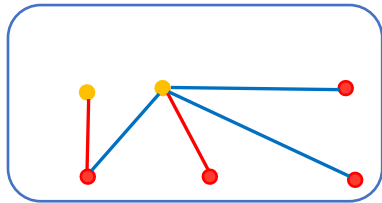


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Communication Model

- How to get $\tilde{O}(n^{1.5})$ communication algorithm?
 - Sample $n^{0.5}$ different **s t** , complexity $n^{1.5}$
 - We use linear sketch to recover the ‘useful’ part of the graph, complexity $n^{1.5}$
- How to get $\tilde{\Omega}(n^{1.5})$ communication lower bound?
 - Reduction from direct sum of set-disjointness